Hicham Alaoui

AI Engineer · LLM systems for finance

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01 Experience

Société Générale CIB (SGCIB) · Casablanca — Feb 2025 – Present

EQD Trading Analyst, index operations and AI tooling — Aug 2025 – Present

- Built a **GitHub Copilot agents toolkit (5 agents, 46 skills)**: methodology PDFs to rule specs, Bloomberg data pulls, reconciliation reports with root-cause analysis; analysts check every output.
- Replaced Excel tracking of index rebalancing with a **7-view Streamlit platform**: exceptions, KPIs, run traces.
- Run monthly and quarterly **rebalancing of equity and derivatives indices** with Structuring and external calculation agents (S&P, Solactive); daily option price, Greeks and parameter checks.

Data Science Intern, Market Access (final-year project) — Feb 2025 – Aug 2025

- Built an **EVT Peaks-over-Threshold model** that recalibrates pre-trade size and price limits from 12M+ trades: **94% precision at a 1.3% false-positive rate**, enabling a **EUR 1.8M RWA release**.
- Shipped it to production: Python/SQL pipelines, Streamlit monitoring, Kupiec backtests, 89% test coverage, Docker, CI; control processing **80% faster**.

gaea21 · NGO, Geneva · Data Scientist Intern — Jun 2024 – Sep 2024

- Modelled CO2 emissions of Swiss agriculture from 50k observations (XGBoost, LSTM, served via a Python API): livestock and fertiliser explain **68% of the variance**.
- Built an LSTM generator that drafted 20+ French articles.

Haut-Commissariat au Plan · Rabat · Data Analyst Intern — Jul 2023 – Aug 2023

- Migrated 1M+ survey records to PostgreSQL: **40% faster queries**.
- Built ARIMA/ARDL poverty indicators and Power BI dashboards.

02 Open-source projects

Methodology RAG assistant — github.com/hicham-alaoui0/methodology-rag

Built an assistant that answers questions over **1,082 pages of index rulebooks** (MSCI, FTSE Russell, STOXX, Solactive) on a 4 GB GPU: hybrid BM25 + bge-m3 search, LLM query rewriting, rank fusion, citation check. **81.9% correct on 94 held-out questions** (95% CI 73–88%); refused 14 of 14 unanswerable questions.

LLM evaluation toolkit — github.com/hicham-alaoui0/llm-evalkit

Wrote a zero-dependency Python library: Wilson intervals, exact McNemar tests, label checks, held-out ledger; CI on Python 3.10, 3.12 and 3.13. Showed that **only 1 of 9 design changes** in the RAG project was a real gain (p = 0.003).

Aloi · a market research agent that checks its figures — hicham-alaoui0.github.io/projects/aloi

Built a hand-written tool loop with 48 tools and 17 playbooks, on Claude, Ollama or any OpenAI-compatible model; every amount and percentage is checked against fetched data. **Passed 27 of 28** tool-use eval questions.

AgentDeck — hicham-alaoui0.github.io/projects/agentdeck

Built a Windows + Android app to answer Claude Code and Codex from one place, with risk rating and an encrypted phone relay. Rust, Tauri; **122 automated tests**.

03 Education

INSEA · Rabat · State Engineer (MSc level) in Applied Economics, Statistics and Big Data — 2022 – 2025

Certificates — Neural Networks and Deep Learning (DeepLearning.AI) · Python Project for Data Science (IBM) · Python for Everybody (University of Michigan) · Prepare Data for Exploration (Google)

04 Skills

AI engineering	LLM agents, tool use, RAG (hybrid search, rerankers), LLM evaluation, Ollama, Claude/OpenAI APIs
ML and statistics	scikit-learn, XGBoost, TensorFlow/Keras, ARIMA/ARDL, extreme value theory, backtesting
Engineering	Python, SQL/PostgreSQL, Streamlit, FastAPI, Docker, GitHub Actions, Git, R, VBA, Power BI
Finance	Index methodology and rebalancing, pre-trade risk limits, option pricing checks, Bloomberg
Languages	Arabic (native), French (C2), English (C1)